National University of Singapore
School of Computing
CS3244: Machine Learning
Solution to Tutorial 5A/5B (2 weeks)

Learning Theory

Readings: *Learning From Data*, Chapter 2 (Training versus Testing)

Part 5A (In-class)

1. **Regularization and l^p norms.** L1 ($\lambda||\mathbf{w}||_1$) and L2 normalization ($\lambda||\mathbf{w}||_2^2$) place different constraints on how C, the budget used for regularization, can be utilized. L1 places a linear constraint over all $\mathbf{w}$, restricting their sum to C; whereas L2 places a squared constraint over all $\mathbf{w}$, restricting the sum of the squares of the weights $\mathbf{w}$ to be C.

   Discuss how the difference in the formulation of the regularizer plays a role in setting the weights for the features. In particular, discuss the tendency of how the two norms differ in their propensity to set a 0 weight for a feature.

   In contrast to L1 norm regularization, L2 norm regularization heavily penalizes larger weights for any feature. L2 thus attempts to pull **all** *weights towards small values. In contrast, L1's linear constraint, may use up its budget C on any set of weights, regardless of its offset from zero.*

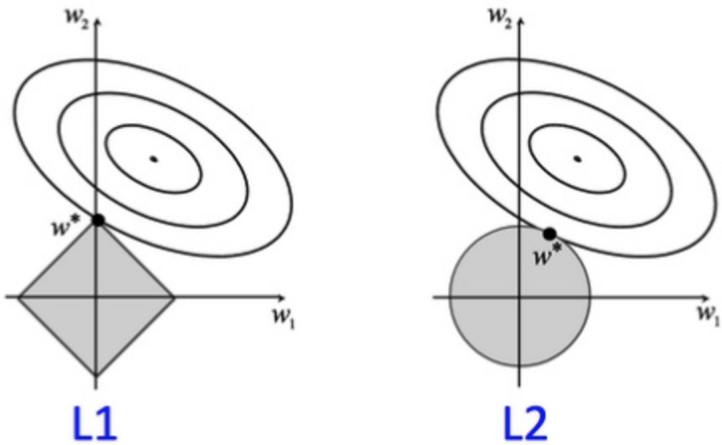


Figure 1: L1 versus L2 norms choosing the best $w = w^*$. Note that due to the topology of the L1 norm, the "points" on the L1 norm provide a larger opportunity to set a component of $\mathbf{w}$ to zero.

   As shown in the diagram above, optimizing $\mathbf{w}$ for minimal E_{in} can often bump into a corner of the L1 norm, causing the weight for the corresponding axis to be set to 0. In L2, since it is a sphere, all points on the L2 budget surface has an equal chance of minimizing E_{in}.

Just for your information, the idea of l^p norms generalizes the concept of norms (and spaces) to encompass both L1 and L2 as part of a more general model where the l^p norm is given by $||x||_p = (|x_1|^p + |x_2|^p + ... + |x_n|^p)^{\frac{1}{p}}$. When $p = 0$, strictly speaking l^0 isn't well defined, but can be taken to mean the number of non-zero entries in $\mathbf{w}$. In this sense L0 "norm" can be thought of the minimizing the number of non-zero weights, similar to the "hard constraint" when we viewed $\mathcal{H}_2$ as a regularized version of $\mathcal{H}_{10}$.

Figure 2: The space of possible regularization outcomes with fixed budget C in under various l^p norms in $\mathbb{R}^3$. Source: https://www.quora.com/What-is-the-difference-between-L1-and-L2-regularization

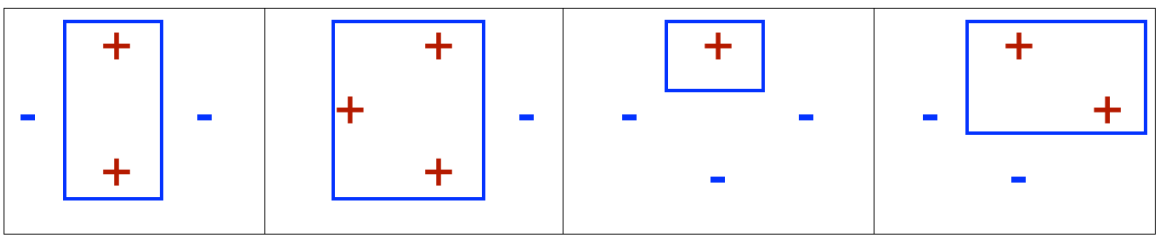
It's also worthwhile knowing that L1 and L2 norm regularization also go by different names in different fields. When studying regression from a statistics perspective, L1 regularization is also known as LASSO by statisticians. L2 regularization, in the same context is known as ridge regression. You can employ more than only form of regularization simultaneously, and when one combines both L1 and L2, this is sometimes referred to as elastic net regularization.

Learning Theory.

1. Show that for the learning model of positive rectangles (aligned horizontally or vertically), $m_{\mathcal{H}}(4) = 2^4$ and $m_{\mathcal{H}}(5) < 2^5$. Hence, give a bound for $m_{\mathcal{H}}(N)$.

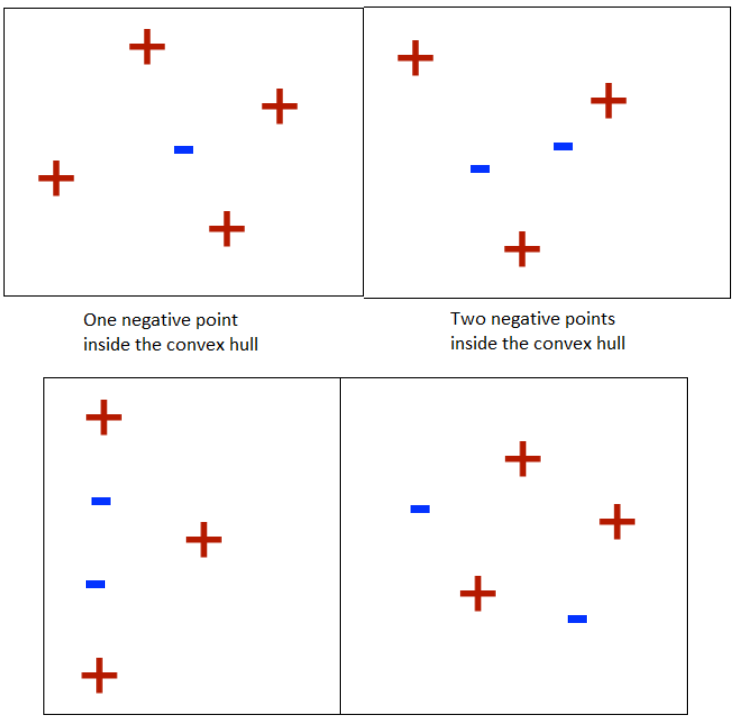
   Remember the growth function, $m_{\mathcal{H}}(N)$ for a hypothesis set $\mathcal{H}$ is defined as the maximum number of dichotomies that can be generated by $\mathcal{H}$ on any set of N points.

   So, we can select a set of four points and check if these points can be shattered by some axis-parallel rectangles. It is quite easy to select and see that these combination of points exist as illustrated in the figure below:



The figure shows 4 possible dichotomies, leaving out two dichotomies (all four points are positive or negative). As we can see that, in this case, we can always find an axis-parallel rectangle to separate out the positive points from the negative points. Hence, $m_{\mathcal{H}}(4) = 2^4$

Now, we need to check different combinations of 5 points. Note that, we can broadly divide the set of combinations as convex sets and non convex sets. It is quite easy to see that we can not find hypotheses that represent a non-convex set or a convex combination of five points that can be shattered with an axis-parallel rectangle:



Another solution to show that 5 points cannot be shattered by axis-parallel rectangles goes by construction. Without loss of generality, we can assume that given a selection of any 5 points, we can assign 4 points (or fewer) with the labels "minimal x value", "minimal y value", "maximum x value" and "maximum y value". With these labels, we form the axis-parallel rectangle to minimally encompass these (up to) four points. By construction, The fifth point must lie in this rectangle, and hence must receive the positive label and thus cannot be assigned the negative label. Hence the set is not shattered, since it cannot represent the dichotomy where the 5th point should be assigned the negative label.

Hence, $m_{\mathcal{H}}(5) < 2^5$*. Now we need to find the upper bound for* $m_{\mathcal{H}}(N)$ *for any* N *points. Here, we apply* ***Theorem 2.4*** *to obtain the upper limit.*

According to the theorem, if $m_{\mathcal{H}}(k) < 2^k$ *for some values of* k*, then*

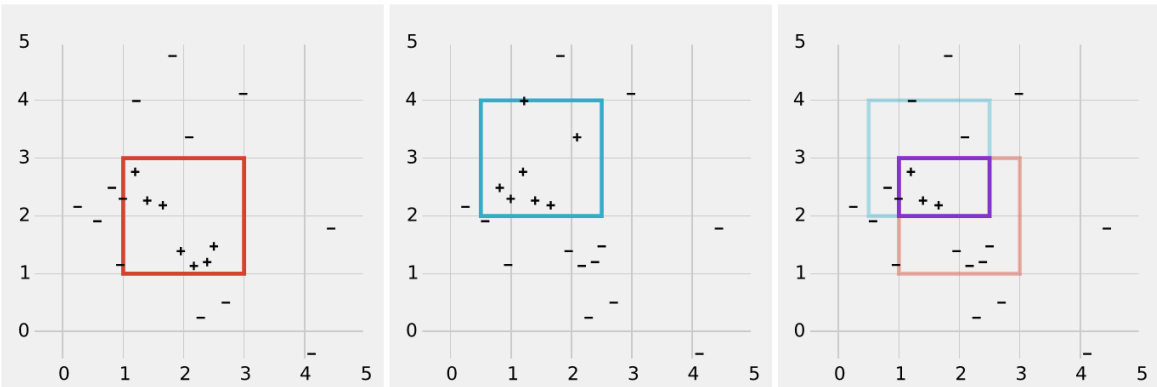
$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^{k-1} {}^nC_k$$

In this case, $k = 5$*. Hence, the upper bound will be* $\sum_{i=0}^{4} {}^nC_5$*.*

2. In this problem, we will explore the hypothesis space where each hypothesis is a combination of two simpler hypotheses. More precisely, given two hypotheses h^1 and h^2, we define $h = h^1 \cap h^2$ as a new hypothesis that labels an example $+1$ only if both h^1 and h^2 give the label $+1$, otherwise, it is labeled -1.

   We can extend this to sets of hypotheses: given two sets of hypotheses $\mathcal{H}^1$ and $\mathcal{H}^2$, define $\mathcal{H}^* = h^1 \cap h^2 : h^1 \in \mathcal{H}^1, h^2 \in \mathcal{H}^2$ as the set of all *intersections of hypothesis pairs* from the two classes $\mathcal{H}^1$ and $\mathcal{H}^2$.

   As an example, let $\mathcal{H}^1$ be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is larger than some threshold a. Let $\mathcal{H}^2$ be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is smaller than some threshold b. Then $\mathcal{H}^*$ would be the set of all intervals (a, b) in $\mathbb{R}$ that assigns $+1$ if the example is inside the interval. Another example is when $\mathcal{H}^1$ and $\mathcal{H}^2$ is the set of all (axis-aligned) squares in $\mathbb{R}^2$, $\mathcal{H}^*$ is the set of all axis-aligned rectangles. This example is illustrated below. On the left, we have a single square classifier h^1; in the middle we again have a square classifier h^2; and on the right, we have $h^1 \cap h^2$, which is a rectangle classifier.



   Suppose that the shattering coefficient of $\mathcal{H}^1$ is $\mathcal{H}^1[n]$ (i.e. the maximum number of ways that the hypothesis space $\mathcal{H}^1$ can label a set of n points is $\mathcal{H}^1[n]$). Similarly, suppose that the shattering coefficient of $\mathcal{H}^2$ is $\mathcal{H}^2[n]$. Show that

$$\mathcal{H}^*[n] \leq \mathcal{H}^1[n]\ \mathcal{H}^2[n].$$

   (Hint: Consider the possible dichotomies that can be generated from these n points for $\mathcal{H}^*$.)

   Given that:

   (a) the maximum possible set of dichotomies that can be generated by hypothesis $\mathcal{H}^1$ from these n data points: $\mathcal{H}^1 = \max_{x_1,\ldots,x_n} |(h(x_1), \ldots, h(x_n))|h \in \mathcal{H}_1|$.

   (b) the maximum possible set of dichotomies that can be generated by hypothesis $\mathcal{H}^2$ from these n data points: $\mathcal{H}^2 = \max_{x_1,\ldots,x_n} |(h(x_1), \ldots, h(x_n))|h \in \mathcal{H}_2|$.

   Now, for $\mathcal{H}^*$, the maximum number of dichotomies, $\mathcal{H}^*$ can be given as follows:

$$\mathcal{H}^* = \max_{x_1,\ldots,x_n} |([h_1(x_1) \cap h_2(x_1)], \ldots, [h_1(x_n) \cap h_2(x_n)])|h_1 \in \mathcal{H}_1,\ h_2 \in \mathcal{H}_2|$$

   by considering all possible conditions. The maximum number cannot exceed $\mathcal{H}_1\mathcal{H}_2$. Hence, $\mathcal{H}^*[n] \leq \mathcal{H}^1[n]\ \mathcal{H}^2[n]$. Note that, equality only holds when each of these dichotomies $([h_1(x_1) \cap$

$h_2(x_1)], ..., [h_1(x_n) \cap h_2(x_n)])$ are different from each other.

Part 5B (Webcast)

1. As forecasted in lecture, prove that the VC dimension of the perceptron in $\mathbb{R}^d$ is $d+1$. *Hint:* You may find it easiest to decompose the proof into the two steps mentioned during lecture: namely that $d_{vc} \leq d+1$ and $d_{vc} \geq d+1$.

   H is the set of hypothesis $w_1x_1 + ... + w_dx_dw_0$, where $w_0, ..., w_d$ are real-valued. Let us first define the definition of convex hull:

   The Convex Hull *of a set of points S is the set of all convex combinations of points S; this is the set of all points that can be written as $\sum_{x_i \in S} \lambda_i x_i$, where $\lambda_i \leq 0$, and $\sum_i \lambda_i = 1$. It is not hard to see that if a half-space has all points from a set S on one side, then the entire convex hull of S must be on that side as well.*

   Lower Bound *Prove that $d_{vc} \geq d+1$ by presenting a set of $d+1$ points in d-dimensional space such that one can partition that set with halfspaces in all possible ways. (And, show how one can partition the set in any desires way.)*

   One good set of $d+1$ points is: The origin and all points with a 1 in one coordinate and zeros in the rest (i.e., all neighbors of the origin on the Boolean cube). Let p_i be the point with a 1 in the ith coordinate. Suppose we wish to partition this set into two pieces S_1 and S_2 with a hyperplane (and, say the origin is in S_1). Then just choose the hyperplane:

   $$\sum_{i:p_i \in S_2} x_i = 1/2$$

   Upper Bound: *In this section, we will apply the following of convex hull to show that the VC-dimension of half-spaces is at most $d+1$.*

   Radon's Theorem: *Let S be a set of $d+2$ points in n dimensions. Then S can be partitioned into two (disjoint) subsets S_1 and S_2 whose convex hulls intersect.*

   If S is a set of $d+2$ points, then by Radon's theorem we may partition S into sets S_1 and S_2 whose convex hulls intersect. Let $p \in S_1$ be a point in that intersection. Assume there exists a hyperplane such that:

   $$\begin{aligned} w \ . \ x_i &\leq w_0, \quad \forall x_i \in S_1 \\ w \ . \ x_i &> w_0, \quad \forall x_i \in S2 \end{aligned}$$

   so that

   $$w \ . \ p \leq w_0$$

which is contradicted by the fact that the point p can also be written as a convex combination of the set of points in S_2. Hence,

$$w \cdot p = \sum_{i:x_i \in S_2} \lambda_i w \cdot x_i > (\sum_{i:x_i \in S_2} \lambda_i) \min_{i:x_i \in S_2} (w \cdot x_i) = \min_{i:x_i \in S_2} (w \cdot x_i) > w_0$$

So no set of $d+2$ points can be shattered and $d_{vc} = d+1$.

2. Show that if the VC dimension of a hypothesis space $\mathcal{H}$ is d, then the growth function $m_{\mathcal{H}}[d]$ is 2^d.

   We refer to the definition of VC dimension to prove this statement. The VC dimension of a hypothesis of a hypothesis space $\mathcal{H}$ is the size of largest set of the points in the instance space that can be shattered by H.

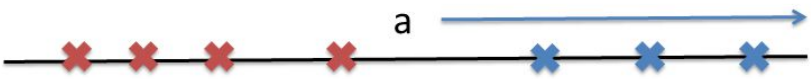
   Hence, there exists a set of d points in the instance space, such that, for each 2^d possible label combinations, we can find a hypothesis $h \in \mathcal{H}$ that can separate the positive points from the negative points. Hence, $m_{\mathcal{H}}[d] = 2^d$.

   Further, we can not find a set of $d+1$ points such that all possible 2^{d+1} dichotomies can be realized by the set $\mathcal{H}$. Otherwise it would contradict $d_{vc} = d$.

3. Compute the maximum number of dichotomies, $m_{\mathcal{H}}(N)$ for the following learning models, and consequently compute the VC dimensions:

   (a) **Rays:** $\mathcal{H}$ contains the functions which are $+1$ on $[a, \infty)$ together with those that are $+1$ on the interval $(-\infty, a]$.

   For the ray hypothesis set,$\mathcal{H}$, each hypothesis can be defined as a single threshold a such that, the all points $p > a$ marked as positive (positive rays) or all points $p < a$ marked as positive (negative rays).



Example of rays

   Note that, given N points, the real line will be split by the points into $N+1$ regions. The dichotomy we get on the N points is decided by which region contains the value of a. Considering only positive rays, we can atmost cover $N+1$ dichotomies. Selecting one region that holds a, all the points right to it will be positive. Similarly, we can cover $N+1$ dichotomies by considering the negative rays. Selecting one region that holds a, all the points right to it will be negative.

Clearly, these two sets have only two dichotomies in common: the case where all N points are marked as positive and the case when all N points are marked as negative. To compute the growth function, we just need to remove these two case. Hence the growth function will be: $m_{\mathcal{H}}(N) = 2N$.

We can easily verify that the VC dimension is 2. We can also compute the VC dimension from the growth function $m_{\mathcal{H}}(N) = 2N$. We just need to find the maximum value of N such that: $2N = 2^N$.

(b) **Intervals:** $\mathcal{H}$ contains the functions which are $+1$ on an interval $[a, b]$ and -1 otherwise together with those that are -1 on an interval $[a, b]$ and $+1$ otherwise.

As before, given N points, the real line will be split by the points into $N+1$ regions. We need two thresholds to select an interval from the real line. Now, the number of ways we can select an interval such that at least one point falls inside the interval is $\binom{N+1}{2}$.

Now for positive intervals, points inside the intervals will be labelled as positive and the remaining points will be labelled as negative. Thus the total number of dichotomies will be covered : $\binom{N+1}{2}$. Similarly for negative intervals, the total number of dichotomies will be covered: $\binom{N+1}{2}$. Note that, all these $2\binom{N+1}{2}$ dichotomies are different from each other.

When all the points are inside the interval, we obtain the case of all positive or negative labels. These cases are selected only once. But the cases with positive labels in front and negative labels in the end – i.e. $\{++++....++--...--\}$ – have been counted twice in both sets. We can easily count the total number of such combinations: $N-1$. Similarly, the cases with negative labels in front and positive labels in the end – i.e. $\{---....--++...++\}$ – has been counted twice. Again the total number of such combinations: $N-1$.

Thus the total number of dichotomies will be: $m_{\mathcal{H}}(N) = 2\binom{N+1}{2} - 2(N-1) = N^2 - N + 2$ As before, we can compute the VC dimension: the maximum value of N such that: $N^2 - N + 2 = 2^N$ i.e. 3.

(c) **Concentric Circles in $\mathbb{R}^d$:** $\mathcal{H}$ contains functions that are $+1$ for $a \leq \sqrt{(x_1^2 + \cdots + x_d^2)} \leq b$.

Note that, given a hypothesis from this set, the points will be labelled as positive or negative only based on their distance from the origin. Clearly, we can convert this problem into a single dimensional one, where each point's coordinate can be replaced with a mapping to complement point in the one dimensional case with an offset equal to its

Euclidean distance from the origin; i.e. $\sqrt{(x_1^2 + \cdots + x_d^2)}$. *With this mapping, all the points will fall in the region* $[0, \infty)$ *on the real line.*
Now, we need to select two thresholds, a and b such that all the points inside the interval will be labelled as positive. Now, this problem is equivalent to our previous question except now the intervals only contains positive labels. So as before, number of dichotomies will be: $\binom{N+1}{2} + 1$. *The additional* 1 *is for the case when we select two thresholds inside the same interval and obtain the dichotomy all points labelled as negative. Hence the growth function* $m_{\mathcal{H}}(N) = \binom{N+1}{2} + 1 = \frac{1}{2}N^2 + \frac{1}{2}N + 1$.

VC dimension: the maximum value of N *such that:* $\frac{1}{2}N^2 + \frac{1}{2}N + 1 = 2^N$ *i.e.* 2.

4. **Hoeffding's Inequality (Hard).** Hoeffding's Inequality is one of the easiest laws of large numbers to prove. Prove Hoeffding's Inequality for Bernoulli random variables, which states:

$$\mathbb{P}[|E_{in}(g) - E_{out}(g)| > \epsilon] \leq 2Me^{-2\epsilon^2 N}$$

You may want to employ Hoeffding's Lemma's in part to prove the inequality. Hoeffding's Lemma states:

LEMMA (HOEFFDINGS LEMMA). Let Z be a Bernoulli random variable with $Z \in [a, b]$. Then, $\mathbb{E}[exp(\lambda(Z - \mathbb{E}[Z]))] \leq exp(\frac{\lambda^2(b-a)^2)}{8})$ for all $\lambda \in \mathbb{R}$.

To prove this result, we will start with a basic of inequality for random variables and slowly build our proof from there[1].

MARKOV'S INEQUALITY: *Let* $Z \geq 0$ *be a non-negative random variable on* $\mathbb{R}$. *Then, for any* $\epsilon > 0$.

$$\begin{aligned}
E(Z) &= \int_{z=0}^{\infty} z.\mathbb{P}(Z = z)dz \\
&= \int_{z=0}^{\epsilon} z.\mathbb{P}(Z = z)dz + \int_{z=\epsilon}^{\infty} z.\mathbb{P}(Z = z)dz \\
&\geq \int_{z=\epsilon}^{\infty} z.\mathbb{P}(Z = z)dz \quad \textit{(removing the first term)} \\
&\geq \epsilon \int_{z=\epsilon}^{\infty} \mathbb{P}(Z = z)dz \\
&= \epsilon\mathbb{P}(Z \geq \epsilon)
\end{aligned}$$

Note that, Z *denotes the random variable and* z *denotes the corresponding value. Here we consider* Z *as a continuous random variable. The same steps hold when* Z *is a discrete random variable. Now, we have* $tP(Z \geq \epsilon) = \mathbb{E}[Z]$.

[1]Concentration inequalities deal with bounding random fluctuations of functions of many independent random variables. The above discussed inequalities are few basic concentration inequalities. If you are interested more in concentration inequalities, please have a look at: `http://videolectures.net/mlss2012_lugosi_concentration_inequalities/`

Rearranging this, we obtain Markov's Inequality:

$$\mathbb{P}(Z \geq \epsilon) = \frac{E(Z)}{\epsilon} \tag{0.0.1}$$

CHEBYSHEV'S INEQUALITY: *Now, consider Z is a random variable on $\mathbb{R}$ with variance, $Var(Z) \leq \infty$. Note that, $P(|Z - E(Z)| \geq \epsilon) = P\Big([Z - E(Z)]^2 \geq \epsilon^2\Big)$. Now, as before $[Z - E(Z)]^2$ is a positive random variable and we choose $\epsilon^2 > 0$. Hence we can apply Markov's Inequality to obtain* CHEBYSHEV'S INEQUALITY, *as follows:*

$$\mathbb{P}(|Z - \mathbb{E}(Z)| \geq \epsilon) = \mathbb{P}\Big([Z - \mathbb{E}(Z)]^2 \geq \epsilon^2\Big) \leq \frac{\mathbb{E}[(Z - \mathbb{E}(Z))^2]}{\epsilon^2} = \frac{Var(Z)}{\epsilon^2}$$

Chebyshev's Inequality is a simple application of Markov's Inequality and will not be used for our proof. We require a more general statement than this for our proof, as in the following:

CHERNOFF BOUNDS: *Again consider Z be any random variable. As we saw, Markov's inequality holds for any non genative random variables. Then for any $\epsilon > 0$,*

$$\mathbb{P}(Z - \mathbb{E}(Z) \geq \epsilon) = \mathbb{P}\Big(e^{\lambda[Z - \mathbb{E}(Z)]} \geq e^{\lambda\epsilon}\Big) \leq \mathbb{E}[e^{\lambda[Z - \mathbb{E}(Z)]}]e^{-\lambda\epsilon}$$

Note that, the Chernoff bound is much stronger than Chebyshevs inequality as it suggests an exponential decaying of the factor $Z - \mathbb{E}(Z)$ as we increase the value of ϵ. The Chernoff bound's use of moment generating functions in an essential way to give exponential deviation bounds.

HOEFFDING'S INEQUALITY: *Consider a hypothesis, $h \in \mathcal{H}$, for each training example, error is always bounded between $[0, 1]$. Let us denote the error for i^{th} training example as Z_i. Then, for N training examples: $E_{in}(g) = \frac{1}{N}\sum_{i=1}^{N} Z_i$. $E_{out}(g)$ represents the expected error for the samples.*

We use Hoeffdings lemma to prove this theorem: giving only the upper tail (i.e., the probability that $E_{in}(g) - E_{out}(g) > \epsilon$) as the lower tail has a similar proof. We use the Chernoff bound technique, which immediately tells us that, for some $\epsilon > 0$:

$$\begin{aligned}
\mathbb{P}(E_{in}(g) - E_{out}(g) > \epsilon) &= \mathbb{P}\Big(\frac{1}{N}\sum_{i=1}^{N}(Z_i - \mathbb{E}(Z_i)) \geq \epsilon\Big) \\
&= \mathbb{P}\Big(\sum_{i=1}^{N}(Z_i - \mathbb{E}(Z_i)) > N\epsilon\Big) \\
&\leq \mathbb{E}\Big[exp\Big(\lambda\sum_{i=1}^{N}[Z_i - \mathbb{E}(Z_i)]\Big)\Big]exp(-\lambda N\epsilon) \\
&= \Big(\prod_{i=1}^{N}\mathbb{E}\Big[exp(\lambda(Z_i - \mathbb{E}[Z_i]))\Big]\Big)exp(-\lambda N\epsilon)
\end{aligned}$$

Note here, Z_i are Bernoulli random variables that can either take 0 or 1. Hence, we can apply the Hoeffding's lemma as follows:

$$\begin{aligned}
\mathbb{P}(E_{in}(g) - E_{out}(g) \geq \epsilon) &\leq \Big(\prod_{i=1}^{N} \mathbb{E}\Big[exp(\lambda(Z_i - \mathbb{E}[Z_i]))\Big]\Big) exp(-\lambda N\epsilon) \\
&= \Big(\prod_{i=1}^{N} exp(\frac{\lambda^2(1-0)^2}{8})\Big) exp(-\lambda N\epsilon) = \Big(\prod_{i=1}^{N} exp(\frac{\lambda}{8})\Big) exp(-\lambda N\epsilon) \\
&= \Big(\prod_{i=1}^{N} exp(\frac{\lambda}{8})\Big) exp(-\lambda N\epsilon)
\end{aligned}$$

Note that the inequality $\mathbb{P}(|E_{in}(g) - E_{out}(g) > \epsilon) \leq \Big(\prod_{i=1}^{N} exp(\frac{\lambda}{8})\Big) exp(-\lambda N\epsilon)$ is true for any value of $\lambda \geq 0$. Hence the following is also true:

$$\mathbb{P}(E_{in}(g) - E_{out}(g) > \epsilon) \leq \min_{\lambda \geq 0} \Big(\prod_{i=1}^{N} exp(\frac{\lambda}{8})\Big) exp(-\lambda N\epsilon)$$

Now we can optimize the expression $\Big(\prod_{i=1}^{N} exp(\frac{\lambda}{8})\Big) exp(-\lambda N\epsilon)$ (i.e, derivative w.r.t. $\lambda = 0$) to obtain the minimum value. The expression of Hoeffding's Inequality can be obtained by replacing the value of λ for which the expression achieves minimum value:

$$\mathbb{P}(E_{in}(g) - E_{out}(g) > \epsilon) \leq \min_{\lambda \geq 0} \Big(\prod_{i=1}^{N} exp(\frac{\lambda}{8})\Big) exp(-\lambda N t) = exp(-2\epsilon^2 N)$$

Symmetrically, the inequality is also valid for another side of the difference i.e.

$$\mathbb{P}(-E_{in}(g) + E_{out}(g) > \epsilon) \leq exp(-2\epsilon^2 N)$$

Note that these are mutually exclusive events (probabilistic events that cannot occur at the same time). Hence, we can add them to obtain the two-sided bound as follows:

$$\mathbb{P}(|E_{in}(g) + E_{out}(g)| > \epsilon) \leq 2exp(-2\epsilon^2 N)$$

Note that, this bound is for only one hypothesis $h \in \mathcal{H}$. Suppose there are total M number of effective hypothesis in the set $\mathcal{H}$, then each of these hypothesis will have same bound. Also each of these hypothesis will produce training and test errors that exclusive of each other. Hence for the hypothesis set $\mathcal{H}$, we can obtain the error bound by adding each of these events for all M hypothesis in the set i.e

$$\mathbb{P}(|E_{in}(g) + E_{out}(g)| > \epsilon) \leq 2M exp(-2\epsilon^2 N).$$